

Color Vision Deficiency

Simulation via VR passthrough



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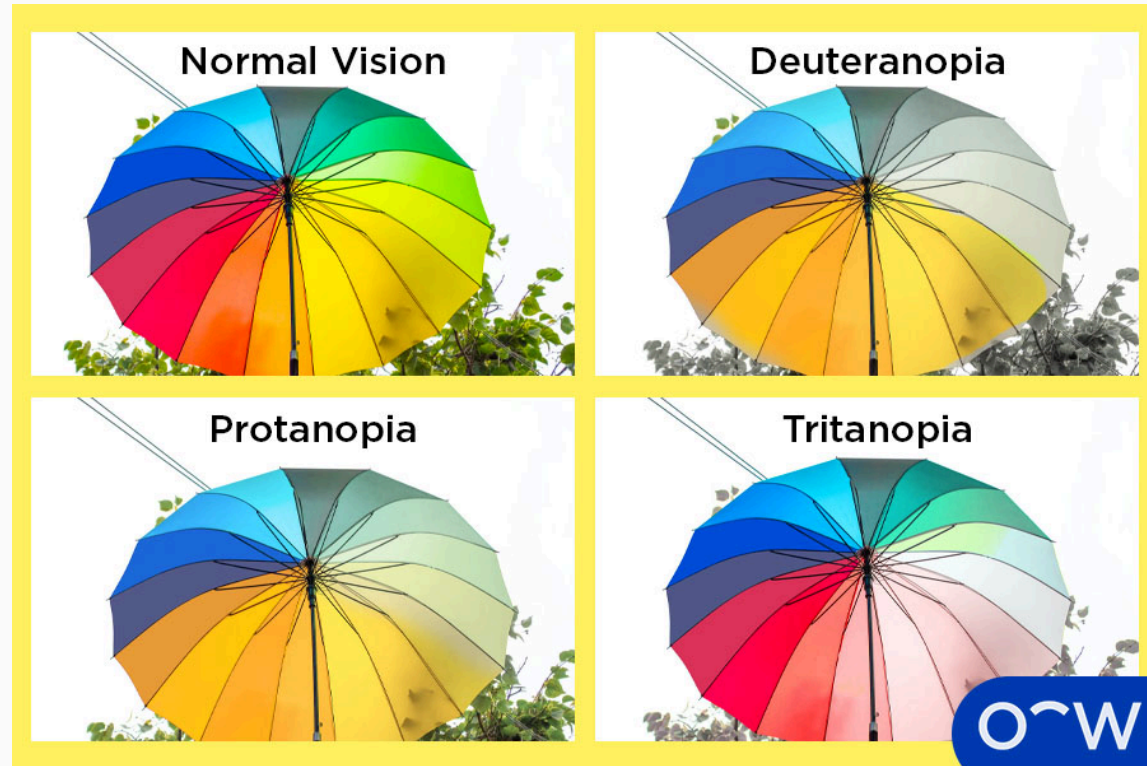
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Color Vision Deficiency

Introduction

Project



[Oscar Wylee Pty Ltd., 2026]

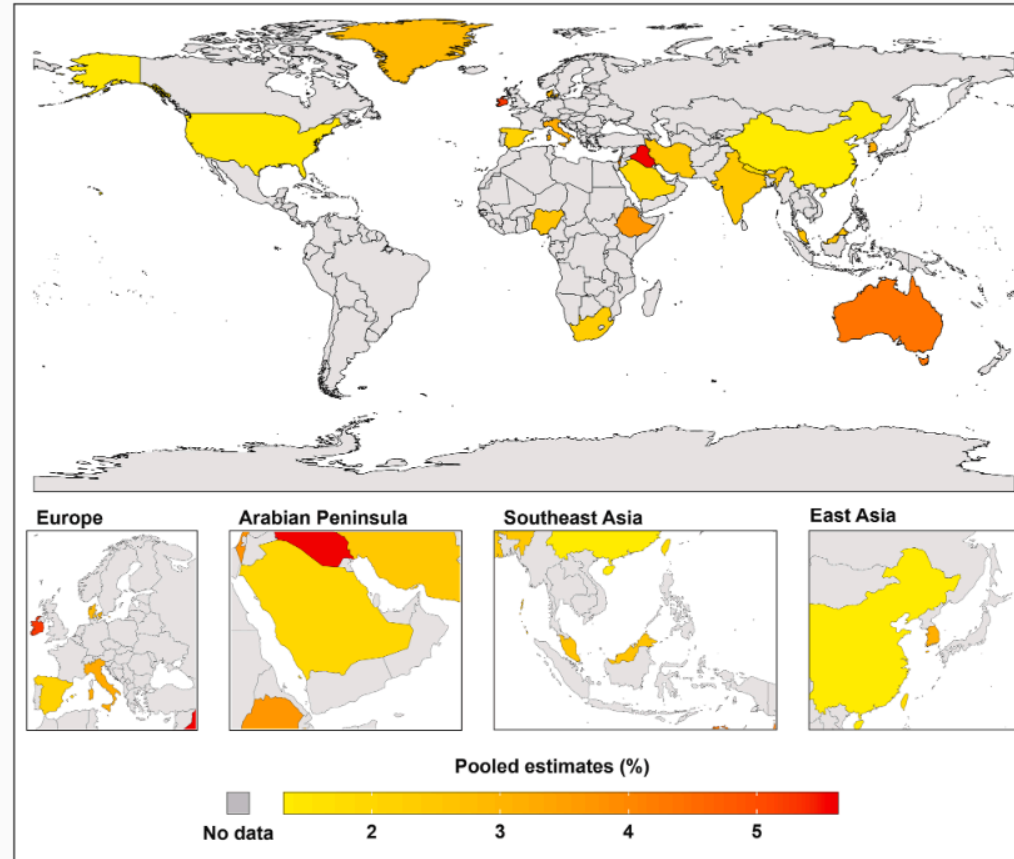


Figure 1: Maps showing global prevalence of congenital color vision deficiency among children and adolescents, 1980 through 2022 [Jeong et al., 2025]

Type	Affected Cones	Prevalence
Protanopia	L (red) absent	1% males
Deuteranopia	M (green) absent	6% males
Tritanopia	S (blue) absent	0.67% males
Achromatopsia	all three absent	NA
Anomalous trichromacy	any impacted	males 1.17%
Dichromacy	two present	males 1.59%
Monochromacy	one present	males 0.36%

[Almustanyir, 2025; Jeong et al., 2025]

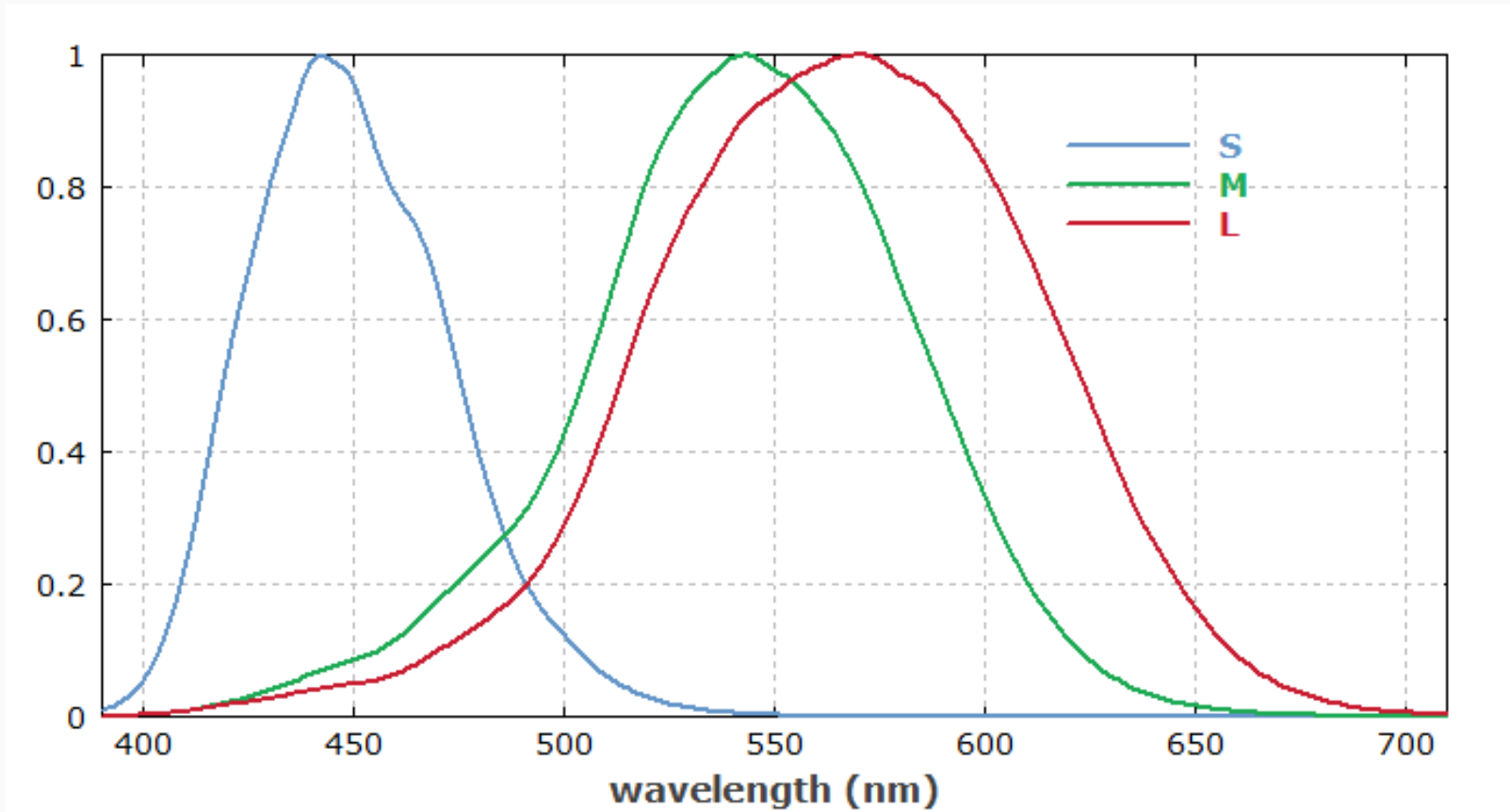
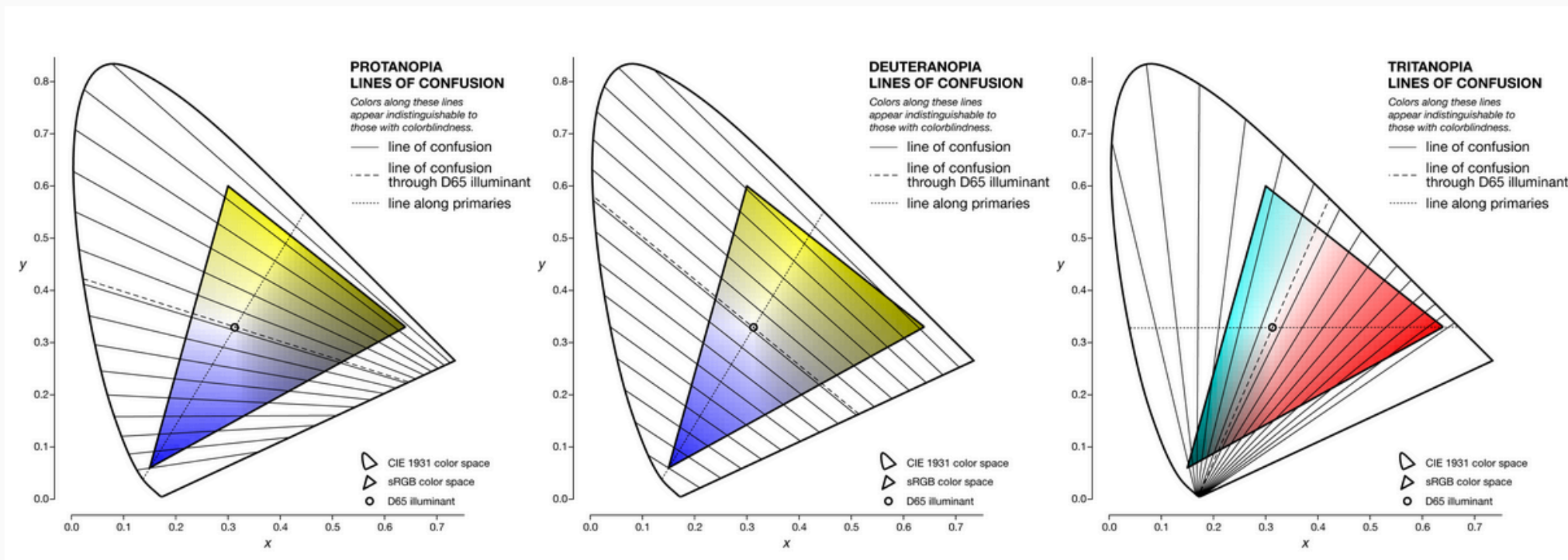
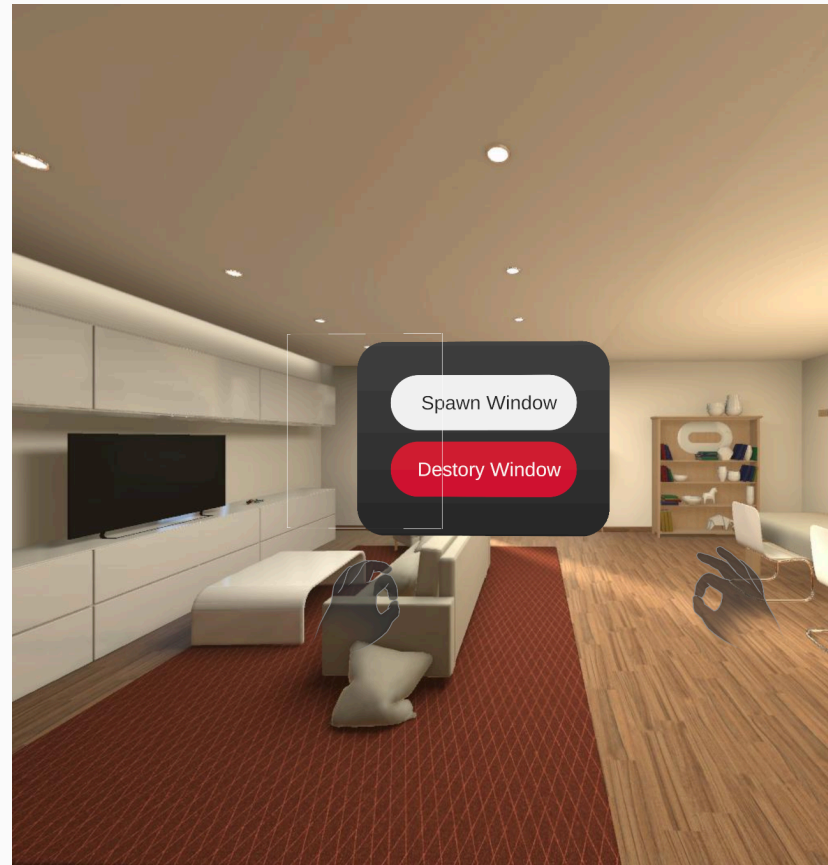
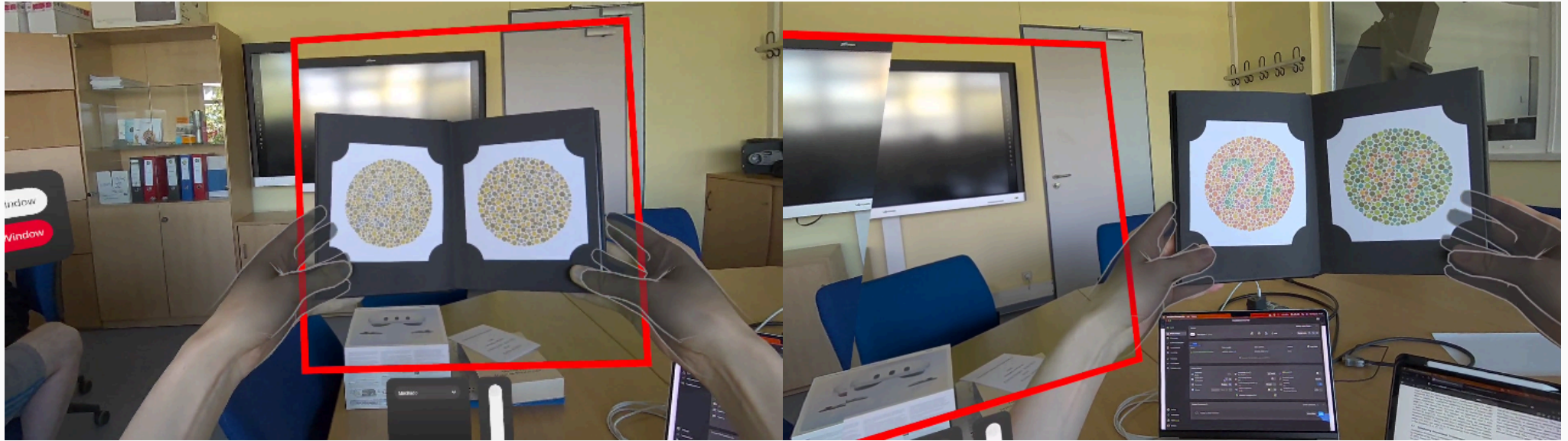


Figure 2: LMS color matching functions from CIE, based on the Stiles and Burch 10° color matching functions. [Paschotta, 2026]



[Krzywinski, 2026]





Demo or Try it yourself

References

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- [Jeong et al., 2025] Jeong, Y. D., Cho, J., Son, Y., Jo, Y., Yim, Y., Kim, T. H., Kim, S., Cho, H., Rahmati, M., Smith, L., Woo, H. G., Kim, J. H., Kim, Y. J., Yang, J. M., and Yon, D. K. (2025). Global Prevalence of Congenital Color Vision Deficiency among Children and Adolescents, 1932–2022. *Ophthalmology*, 132(12), 1431–1444.
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- [Paschotta, 2026] Paschotta, D. R. (2026). Color Spaces.